Freight Rate Prediction Challenge

Model approach, data quality, validation, and submission artifacts

1. Objective and data

The objective is to predict posted freight rates for 12,000 unlabeled loads and produce a fixed-input daily prediction series for December 2025. The labeled development dataset contains 48,000 rows from January 1 through October 31, 2025. The final prediction set covers November and December 2025.

2. Data-quality review

Check	Labeled data (48,000 rows)	Validation data (12,000 rows)
Missing weight	300	165
Zero/negative weight	292	145
Missing market_index	374	249
Duplicate load_id	0	0

Treatment: zero/negative weights are treated as missing and filled with the median weight from the relevant training data. Missing numeric feature values are median-imputed using training-only statistics. Missing categorical values are mapped to “Unknown”. The load_id identifier is excluded from model features. Rows are not removed solely because an input feature is missing.

3. Feature engineering and model

CatBoostRegressor is used as a nonlinear regression model. Features include pickup, delivery, route (pickup-to-delivery), equipment, pickup/delivery coordinates, distance, weight, market_index, quote_signal, and year/month/day/day-of-week/day-of-year extracted from date. The target is posted_rate. CatBoost was selected for its ability to model nonlinear interactions and categorical route/equipment variables with limited preprocessing.

4. Validation design and measured results

A chronological holdout was used: January–September 2025 (43,147 rows) for development and October 2025 (4,853 rows) for holdout validation. This avoids randomly mixing earlier and later dates and better simulates predicting future loads. Early stopping selected 163 boosting iterations. The final model was then retrained on all 48,000 labeled rows before predicting the supplied validation and December inputs.

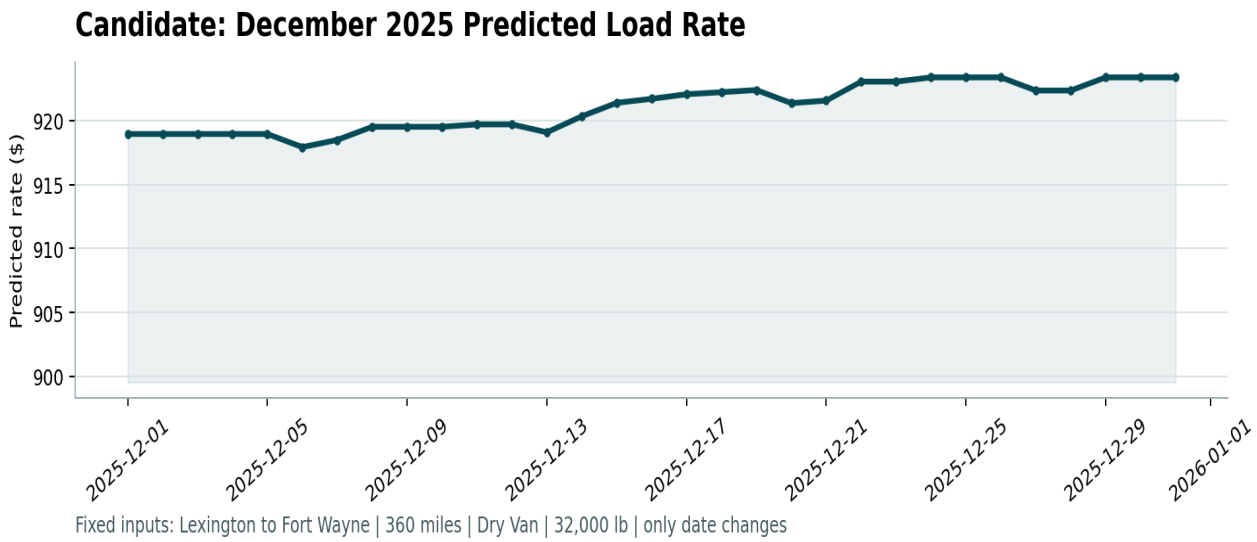
Metric	October holdout result
MAE	$108.06
RMSE	$647.14
R^2	0.8208
Mean absolute percentage error	5.58%

Interpretation: MAE is the average absolute dollar error on October rows. RMSE is more sensitive to large misses; its higher value suggests some large errors/outliers. These are local holdout metrics, not Spotter’s final hidden-validation score.

5. Outputs and verification

The solution generated outputs/validation_predictions.csv with exactly two columns (load_id,predicted_rate) and 12,000 rows. It generated 31 December predictions using the fixed Lexington-to-Fort-Wayne route, 360 miles, Dry Van, and 32,000 lb, with date as the changing input. The supplied scorer ran successfully: it validated all 12,000 IDs and the 31 December dates and produced scorer_results/candidate_december.png.

6. December predicted rate chart



The chart is generated by the supplied score.py from the model's December predictions. Predicted rates range from approximately $917.92 to $923.37 across the 31 days.

7. Reproducibility and limitations

From the project root, install dependencies with pip install -r requirements.txt, run python train_model.py --data-dir data --output-dir outputs, then run python score.py --predictions outputs/validation_predictions.csv --december-predictions outputs/december_chart_inputs.csv --output-dir scorer_results. The model is a baseline and can be improved through residual analysis, route-level checks, tuning, and comparison against other algorithms. The hidden final evaluation metric is not available locally.